

Machine learning notes

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1 Machine Learning Fundamentals

1.1 What is Machine Learning

Machine learning (ML) is a subset of Artificial Intelligence (AI) where the system automatically learns pattern from data and improves their performance without being explicitly programmed.

1.2 Supervised learning vs Unsupervised learning

Supervised learning trains models using labeled data to predict outcome.
Unsupervised learning finds hidden patterns and structures in unlabeled data.

1.3 Classification and regression

Classification predicts discrete categories or labels (e.g. spam vs. not spam).
Regression predicts continuous numerical values (e.g. predicting house prices)

1.4 Train / validation / test split

training set: used to train the model.
validation set: during training to tune hyper-parameters and prevent overfitting.
testing set: used only once for evaluation of the final model.

1.5 Overfitting vs Underfitting

overfitting: The model is too complex. It learns the training data very well, but fails on the testing/evaluation data.
underfitting: The model is too simple to understand the data, so it performs poorly on both the data it was trained on and the new data it is tested on.

1.6 Basic data preprocessing

- a) missing value - drop, imputation
- b) categorical data - label encoding, one hot encoding

c) feature scaling - normalization, standardization

1.normalization(min-max)

2.standardization(z-score)

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

$$X_{\text{std}} = \frac{X - \mu}{\sigma} \quad (2)$$

d) outliers - z-score, IQR